

N=32 ICON-YAC-CLEO: compact and ragged outputs are independent observer paths

Source traced in the exact CLEO bubble code used on Levante; 241 stored states at a 30 s observation interval.

OBSTEP = 30 s

Live CLEO model state at an observation time

d_supers: superdroplet attributes in memory
(id, gridbox membership, coordinates, multiplicity ξ , radius r , solute mass)
d_gbxs: thermodynamics and wind state in memory

same live state

same live state

Compact observer path — direct in-memory reductions

MassMomentsObserver: Kokkos reduction separately in every gridbox
 $M_0 = \sum \xi$; $M_1 = \sum \xi m$; $M_2 = \sum \xi m^2$, using each live drop.mass()
StateObserver: live T , p , q_v , u , v , w gridbox state
MonitorPrecipitationObserver: live boundary-crossing monitor

No saved ragged array is read in this calculation.

Ragged particle-output path — attribute copies then serialization

SuperdropsObserver: loops over all live d_supers and copies attributes
sdld, coord1/2/3, ξ , r , msol $\rightarrow$ one concatenated 'superdroplets' axis
raggedcount(time) marks the number of records in each saved state

sdgbxindex was commented out in this bubble executable.

Compact arrays stored in bubble3d_sol.zarr

massmom0/1/2, massmom*_raindrops, temp, press, qvap, uvel, vvel, wvel, precip
Each dense field: (time, gbindex) = (241, 768)
These are the appropriate source for gridded mass, thermodynamic, wind, and boundary-flux diagnostics.

Ragged arrays stored in the same Zarr group

sdld, coord1/2/3, ξ , radius, msol: each (superdroplets) = (5,339,479)
raggedcount and totnsupers: each (time) = (241,)
Useful for particle snapshots, but any reconstructed bulk statistic must agree with the compact observer before use.

N=32 all-time consistency result and inference

The audit uses CLEO's total-mass definition: $m = 4\pi\rho r^3/3 + m_{\text{sol}}(1 - \rho_l/\rho_{\text{sol}})$.
At $t = 0$: raw/compact $M_1 = 0.99999994$. At 59 min: raw/compact $M_1 = 1.9007$ while water-only/compact = 1.9001;
the aerosol term is only 0.03% of raw mass. Aerosol accounting therefore cannot explain the discrepancy.
The mismatch proves separate output paths, makes raw-derived bulk diagnostics provisional, and does not itself identify the cause.